Assignment IV - Discrete Math(H)

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Q.1

> Q.1 Use induction to prove that 4 divides $2n^2 + 6n$ whenever n is a positive integer.

Base case: Let $n = 1$. Then

$$2 \cdot 1^2 + 6 \cdot 1 = 2 + 6 = 8$$

Then $4 \mid 8$.

Induction: Assume that for some positive integer k,

$$4 \mid (2k^2 + 6k).$$

Hence there exists an integer m such that

$$2k^2 + 6k = 4m.$$

Inductive step:

$$\begin{aligned} 2(k+1)^2 + 6(k+1) &= 2(k^2 + 2k + 1) + 6k + 6 \\ &= 2k^2 + 4k + 2 + 6k + 6 \\ &= 2k^2 + 10k + 8 \\ &= (2k^2 + 6k) + 4k + 8. \end{aligned}$$

We have $2k^2 + 6k = 4m$. Substituting, we obtain

$$\begin{aligned} 2(k+1)^2 + 6(k+1) &= 4m + 4k + 8 \\ &= 4m + 4(k+2) \\ &= 4(m+k+2). \end{aligned}$$

Therefore

$$4 \mid \left(2(k+1)^2 + 6(k+1)\right),$$

Hence, for all positive integers n, the integer $2n^2 + 6n$ is divisible by 4.

Q.2

Q.2 Prove that if $A_1, A_2, \ldots, A_n$ and B are sets, then

$$\begin{aligned} &(A_1 - B) \cap (A_2 - B) \cap \cdots \cap (A_n - B) \\ = \quad &(A_1 \cap A_2 \cap \cdots \cap A_n) - B. \end{aligned}$$

Left side: For every i, we have $x \in A_i$, so

$$x \in A_1 \cap A_2 \cap \cdots \cap A_n.$$

At the same time, we also have $x \notin B$. Hence

$$x \in (A_1 \cap A_2 \cap \cdots \cap A_n) - B.$$

Therefore,

$$(A_1 - B) \cap \cdots \cap (A_n - B) \subseteq (A_1 \cap \cdots \cap A_n) - B$$

Right side:

$$x \in A_1 \cap A_2 \cap \cdots \cap A_n \quad \wedge \quad x \notin B.$$

$x \in A_1 \cap A_2 \cap \cdots \cap A_n$ is equivalent to for each $i = 1, \ldots, n$, we have $x \in A_i$.

Therefore, for every i we have $x \in A_i$ and $x \notin B$.

Hence,

$$(A_1 \cap \cdots \cap A_n) - B \subseteq (A_1 - B) \cap \cdots \cap (A_n - B).$$

Overall, we have:

$$(A_1 \cap \cdots \cap A_n) - B = (A_1 - B) \cap \cdots \cap (A_n - B).$$

`Q.E.D.`.

Q.3

Q.3 Prove that if $h > -1$, then $1 + nh \leq (1+h)^n$ for all nonnegative integers n. This is called **Bernoulli's inequality**.

PF:

Base case: Let $n = 0$. Then

$$1 + 0 \cdot h = 1 = (1+h)^0.$$

So the inequality holds when $n = 0$.

Induction: Assume that for some nonnegative integer k,

$$1 + kh \leq (1+h)^k.$$

Inductive step:

Since $h > -1$, we have $1 + h > 0$. Multiplying both sides of the induction hypothesis by $1 + h$ (a positive number), we obtain

$$(1+kh)(1+h) \leq (1+h)^k(1+h) = (1+h)^{k+1}$$

Now expand the left-hand side:

$$\begin{aligned}(1+kh)(1+h) &= 1 + h + kh + kh^2 \\ &= 1 + (k+1)h + kh^2\end{aligned}$$

Because $k \geq 0$ and $h^2 \geq 0$, we have $kh^2 \geq 0$, so

$$1 + (k+1)h \leq 1 + (k+1)h + kh^2$$

Then

$$1 + (k+1)h \leq 1 + (k+1)h + kh^2 = (1+kh)(1+h) \leq (1+h)^{k+1}$$

Therefore

$$1 + (k+1)h \leq (1+h)^{k+1}$$

Hence, for all nonnegative integers n,

$$1 + nh \leq (1+h)^n$$

`Q.E.D.`.

Q.4

Q.4 Let $P(n)$ be the statement that a postage of n cents can be formed using just 4-cent stamps and 7-cent stamps. The parts of this exercise outline a strong induction proof that $P(n)$ is true for $n \geq 18$.

(a) Show statements $P(18)$, $P(19)$, $P(20)$ and $P(21)$ are true, completing the basis step of the proof.

(b) What is the inductive hypothesis of the proof?

(c) What do you need to prove in the inductive step?

(d) Complete the inductive step for $k \geq 21$.

(e) Explain why these steps show that this statement is true whenever $n \geq 18$.

Sol:

1. Basis step:

For 18 cents:

$$18 = 4 + 7 + 7,$$

so $P(18)$ holds.

For 19 cents:

$$19 = 4 + 4 + 4 + 7,$$

so $P(19)$ holds.

For 20 cents:

$$20 = 4 + 4 + 4 + 4 + 4,$$

so $P(20)$ holds.

For 21 cents:

$$21 = 7 + 7 + 7,$$

so $P(21)$ holds.

Thus the basis step is complete.

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2. Assume that for some integer $k \geq 21$, the statement $P(n)$ is true for every integer n with

$$18 \leq n \leq k.$$

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3. We need to prove that $P(k+1)$ is also true.

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4. Let $k \geq 21$ and assume that $P(n)$ holds for all integers n with $18 \leq n \leq k$.

To show that $P(k+1)$ holds. Since $k \geq 21$, we have

$$k + 1 \geq 22 \quad \text{and hence} \quad k + 1 - 4 = k - 3 \geq 18.$$

By the inductive hypothesis, $P(k-3)$ is true, and we can add one more 4-cent stamp.

Therefore, a postage of $k+1$ cents can be formed using only 4-cent and 7-cent stamps, and thus $P(k+1)$ holds.

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5. We have shown that:
    1. $P(18)$, $P(19)$, $P(20)$, and $P(21)$ are true.
    2. For any integer $k \geq 21$, if $P(n)$ is true for every integer n with $18 \leq n \leq k$, then $P(k+1)$ is also true.

By strong mathematical induction, it's easy to prove.

Q.5

Q.5 Show that the principle of mathematical induction and strong induction are equivalent. That is, each can be shown to be valid from the other.

PF:

Assume the ordinary induction principle is valid.

Define

$$Q(n) : P(1) \wedge P(2) \wedge \cdots \wedge P(n).$$

- Base case: $Q(1)$ is just $P(1)$.
- Induction step: assume $Q(k)$, i.e. $P(1), \ldots, P(k)$ are all true.

  Then $P(k+1)$ is true, so

  $$Q(k+1) = Q(k) \wedge P(k+1)$$

  is true.

By ordinary induction on $Q(n)$ we get $Q(n)$ for all $n \geq 1$.

Hence each $P(n)$ is true, so the strong induction conclusion holds.

Assume the strong induction principle is valid.

- Base case: $P(1)$ is true.
- Strong step: assume $P(1), \ldots, P(k)$ are all true. In particular, $P(k)$ is true, so we obtain

$P(k+1)$.

by strong induction $P(n)$ holds for all $n \geq 1$.

Therefore, ordinary induction and strong induction are equivalent.

`Q.E.D.`.

Q.6

Q.6 Suppose that the function f satisfies the recurrence relation $f(n) = 2f(\sqrt{n}) + \log n$ whenever n is a perfect square greater than 1 and $f(2) = 1$.

(a) Find $f(16)$

(b) Find a big-O estimate for $f(n)$. [Hint: make the substitution $m = \log n$.]

Sol:

1.

$$f(4) = 2f(2) + \log 4 = 2 \cdot 1 + \log 4 = 2 + \log 4.$$

$$\begin{aligned} f(16) &= 2f(4) + \log 16 \\ &= 2(2 + \log 4) + \log 16 \\ &= 4 + 2\log 4 + \log 16. \end{aligned}$$

If log is base 2, then $\log 4 = 2$ and $\log 16 = 4$, so

$$f(16) = 4 + 2 \cdot 2 + 4 = 12.$$

2.

Let $m = \log n$ and define

$$g(m) = f(2^m).$$

Then

$$g(m) = 2g(m/2) + m, \qquad g(1) = 1.$$

The recurrence

$$T(m) = 2T(m/2) + m$$

By the Master Theorem, we have:

$$g(m) = \Theta(m \log m).$$

Therefore

$$f(n) = g(\log n) = \Theta\big((\log n)(\log\log n)\big),$$

so

$$f(n) = O\big((\log n)(\log\log n)\big).$$

Q.7

Q.7 The running time of an algorithm A is described by the following recurrence relation:

$$S(n) = \begin{cases} b & n = 1 \\ 9S(n/2) + n^2 & n > 1 \end{cases}$$

where b is a positive constant and n is a power of 2. The running time of a competing algorithm B is described by the following recurrence relation:

$$T(n) = \begin{cases} c & n = 1 \\ aT(n/4) + n^2 & n > 1 \end{cases}$$

where a and c are positive constants and n is a power of 4. For the rest of this problem, you may assume that n is always a power of 4. You should also assume that $a > 16$. (Hint: you may use the equation $a^{\log_2 n} = n^{\log_2 a}$)

(a) Find a solution for $S(n)$. Your solution should be in *closed form* (in terms of b if necessary) and should *not* use summation.

(b) Find a solution for $T(n)$. Your solution should be in *closed form* (in terms of a and c if necessary) and should *not* use summation.

(c) For what range of values of $a > 16$ is Algorithm B at least as efficient as Algorithm A asymptotically ($T(n) = O(S(n))$)?

1.

Let $n = 2^k$ and define $u_k = S(2^k)$. Then

$$u_0 = S(1) = b, \qquad u_k = 9u_{k-1} + (2^k)^2 = 9u_{k-1} + 4^k.$$

So

$$u_k - 9u_{k-1} = 4^k.$$

Homogeneous solution: $C9^k$. Try a particular solution $A4^k$:

$$A4^k - 9A4^{k-1} = A4^{k-1}(4 - 9) = 4^k \Rightarrow A = -\frac{4}{5}.$$

Hence

$$u_k = C9^k - \frac{4}{5}4^k.$$

Using $u_0 = b$:

$$b = C - \frac{4}{5} \Rightarrow C = b + \frac{4}{5}.$$

Since $n = 2^k$ and $9^k = n^{\log_2 9}$, $4^k = n^2$, we get

$$S(n) = \left(b + \frac{4}{5}\right)n^{\log_2 9} - \frac{4}{5}n^2 = \Theta\left(n^{\log_2 9}\right).$$

2.

Let $n = 4^k$ and define $v_k = T(4^k)$. Then

$$v_0 = T(1) = c, \qquad v_k = av_{k-1} + (4^k)^2 = av_{k-1} + 16^k.$$

So

$$v_k - av_{k-1} = 16^k.$$

Homogeneous solution: Ca^k. Try a particular solution $B16^k$:

$$B16^k - aB16^{k-1} = B16^{k-1}(16 - a) = 16^k \Rightarrow B = \frac{16}{16 - a} \quad (a \neq 16).$$

Thus

$$v_k = Ca^k + \frac{16}{16 - a}16^k.$$

Using $v_0 = c$:

$$c = C + \frac{16}{16 - a} \Rightarrow C = c - \frac{16}{16 - a}.$$

Since $n = 4^k$, we have $a^k = n^{\log_4 a}$ and $16^k = n^2$, so

$$T(n) = \left(c - \frac{16}{16 - a}\right)n^{\log_4 a} + \frac{16}{16 - a}n^2 = \Theta\left(n^{\log_4 a}\right) \quad (a > 16).$$

3.

Asymptotically,

$$S(n) = \Theta\big(n^{\log_2 9}\big), \qquad T(n) = \Theta\big(n^{\log_4 a}\big).$$

We have $T(n) = O(S(n))$ iff

$$\log_4 a \le \log_2 9 \iff \frac{1}{2}\log_2 a \le \log_2 9 \iff \log_2 a \le \log_2 81 \iff a \le 81.$$

With the assumption $a > 16$, Algorithm B is at least as efficient as Algorithm A when

$$16 < a \le 81.$$

Q.8

Q.8 Consider three subsets A, B, C of a set S.

(1) Write a formula of $|\overline{A} \cap \overline{B} \cap \overline{C}|$ using the inclusion-exclusion principle.

(2) Use the formula in (1) to count the number of integers from 1 to 1000 (inclusive) which are not multiples of 10, 4 or 15.

1.

By De Morgan's law,

$$\overline{A} \cap \overline{B} \cap \overline{C} = \overline{A \cup B \cup C}.$$

Hence

$$|\overline{A} \cap \overline{B} \cap \overline{C}| = |S| - |A \cup B \cup C|.$$

Using inclusion–exclusion for three sets,

$$|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |A \cap C| - |B \cap C| + |A \cap B \cap C|.$$

So

$$|\overline{A} \cap \overline{B} \cap \overline{C}| = |S| - \Big(|A| + |B| + |C| - |A \cap B| - |A \cap C| - |B \cap C| + |A \cap B \cap C|\Big).$$

2.

Let $S = \{1, 2, \ldots, 1000\}$, so $|S| = 1000$.

Let

- A be the multiples of 10,
- B be the multiples of 4,
- C be the multiples of 15.

Then

$$|A| = \left\lfloor \frac{1000}{10} \right\rfloor = 100, \quad |B| = \left\lfloor \frac{1000}{4} \right\rfloor = 250, \quad |C| = \left\lfloor \frac{1000}{15} \right\rfloor = 66.$$

Intersections (using least common multiples):

$$|A \cap B| = \left\lfloor \frac{1000}{20} \right\rfloor = 50, \quad |A \cap C| = \left\lfloor \frac{1000}{30} \right\rfloor = 33, \quad |B \cap C| = \left\lfloor \frac{1000}{60} \right\rfloor = 16,$$

and

$$|A \cap B \cap C| = \left\lfloor \frac{1000}{60} \right\rfloor = 16.$$

Thus

$$\begin{aligned} |A \cup B \cup C| &= 100 + 250 + 66 - 50 - 33 - 16 + 16 \\ &= 333. \end{aligned}$$

The required number is

$$|\overline{A} \cap \overline{B} \cap \overline{C}| = |S| - |A \cup B \cup C| = 1000 - 333 = 667.$$

So there are 667 integers from 1 to 1000 (inclusive) that are not multiples of 10, 4, or 15.

Q.9

Q.9 Suppose that $n \geq 1$ is an integer.

(a) How many functions are there from the set $\{1, 2, \ldots, n\}$ to the set $\{1, 2, 3\}$?

(b) How many of the functions in part (a) are one-to-one functions?

(c) How many of the functions in part (a) are onto functions?

Sol:

1.

The domain has n elements and the codomain has 3 elements.

Each of the n elements can be mapped independently to any of 3 values, so the total number of functions is 3^n.

2.

A function from $\{1, \ldots, n\}$ to $\{1, 2, 3\}$ can be injective only if $n \leq 3$.

- If $n > 3$, there are no one-to-one functions (answer 0).
- If $1 \leq n \leq 3$, an injective function corresponds to a permutation of n distinct images chosen from 3 elements:

$$P(3, n) = \frac{3!}{(3-n)!}.$$

3.

A function can be onto only if $n \geq 3$.

- If $n < 3$, there are no onto functions (answer 0).

Assume $n \geq 3$. The total number of functions is 3^n.

Use inclusion–exclusion to subtract those missing at least one value.

Let A_i be the set of functions that never take the value i.

Then

- $|A_i| = 2^n$ for each i (only two values allowed),
- $|A_i \cap A_j| = 1^n = 1$ for each pair $i \neq j$,
- $|A_1 \cap A_2 \cap A_3| = 0$.

Hence the number of non-onto functions is

$$|A_1 \cup A_2 \cup A_3| = 3 \cdot 2^n - 3 \cdot 1^n + 0 = 3 \cdot 2^n - 3.$$

Therefore the number of onto functions is

$$3^n - (3 \cdot 2^n - 3) = 3^n - 3 \cdot 2^n + 3 \qquad (n \geq 3).$$

Q.10

Q.10 Prove that the binomial coefficient

$$\binom{240}{120}$$

is divisible by $242 = 2 \cdot 121$.

PF:

Write

$$\binom{240}{120} = \frac{240!}{120!\,120!}.$$

For a prime p, the exponent of p in $n!$ is

$$v_p(n!) = \left\lfloor \frac{n}{p} \right\rfloor + \left\lfloor \frac{n}{p^2} \right\rfloor + \left\lfloor \frac{n}{p^3} \right\rfloor + \cdots,$$

so

$$v_p\left(\binom{240}{120}\right) = v_p(240!) - 2v_p(120!).$$

Exponent of 11:

$$v_{11}(240!) = \left\lfloor \frac{240}{11} \right\rfloor + \left\lfloor \frac{240}{11^2} \right\rfloor = 21 + 1 = 22,$$

$$v_{11}(120!) = \left\lfloor \frac{120}{11} \right\rfloor + \left\lfloor \frac{120}{11^2} \right\rfloor = 10 + 0 = 10.$$

Hence

$$v_{11}\left(\binom{240}{120}\right) = 22 - 2 \cdot 10 = 2.$$

Thus $\binom{240}{120}$ is divisible by $11^2 = 121$.

Exponent of 2.:

$$\begin{aligned} v_2(240!) &= \left\lfloor \frac{240}{2} \right\rfloor + \left\lfloor \frac{240}{4} \right\rfloor + \left\lfloor \frac{240}{8} \right\rfloor + \left\lfloor \frac{240}{16} \right\rfloor + \left\lfloor \frac{240}{32} \right\rfloor + \left\lfloor \frac{240}{64} \right\rfloor + \left\lfloor \frac{240}{128} \right\rfloor \\ &= 120 + 60 + 30 + 15 + 7 + 3 + 1 = 236, \end{aligned}$$

$$v_2(120!) = 60 + 30 + 15 + 7 + 3 + 1 = 116.$$

So

$$v_2\left(\binom{240}{120}\right) = 236 - 2 \cdot 116 = 4.$$

Hence $\binom{240}{120}$ is divisible by 2^4, in particular by 2.

Combining, $\binom{240}{120}$ is divisible by $2 \cdot 11^2 = 242$, as required.

`Q.E.D.`.

Q.11

Q.11 Consider all permutations of the letters A, B, C, D, E, F, G.

(a) How many of these permutations contains the strings ABC and DE (each as consecutive substring)?

(b) In how many permutations does A precede B? (not necessary immediately)

Sol:

1. Treat the block ABC as one letter $[ABC]$ and the block DE as another letter $[DE]$. Together with F and G, we have four
   distinct objects:

$$[ABC],\ [DE],\ F,\ G.$$

Then:

$$4! = 24$$

Hence the number of permutations containing ABC and DE as consecutive substrings is 24.

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2. There are in total

$$7! = 5040$$

permutations of the seven letters.

For any permutation, if we swap the positions of A and B, we obtain a different permutation. In each such pair, exactly one permutation has A before B, and the other has B before A. This gives a pairing of all permutations into two-element sets of this form.

Then the number is

$$\frac{7!}{2} = 2520.$$

Q.12

Q.12 Consider the equation

$$x_1 + x_2 + x_3 + x_4 + x_5 = 10.$$

with five variables.

(1) Count the number of integer solutions, with $x_1 \geq 3$, $x_2 \geq 0$, $x_3 \geq -2$, $x_4 \geq 0$, and $x_5 \geq 0$.

(2) Count the number of integer solutions, with $0 \leq x_1 \leq 5$ and $x_2, x_3, x_4, x_5 \geq 0$.

Sol:

1.

Set

$$y_1 = x_1 - 3, \quad y_2 = x_2, \quad y_3 = x_3 + 2, \quad y_4 = x_4, \quad y_5 = x_5.$$

Then $y_i \geq 0$ and

$$(y_1 + 3) + y_2 + (y_3 - 2) + y_4 + y_5 = 10 \iff y_1 + y_2 + y_3 + y_4 + y_5 = 9.$$

The number of nonnegative integer solutions of $y_1 + \cdots + y_5 = 9$ is, by stars and bars,

$$\binom{9+5-1}{5-1} = \binom{13}{4} = 715.$$

So the answer is 715.

2.

First count all nonnegative solutions (ignoring $x_1 \leq 5$):

$$\binom{10+5-1}{5-1} = \binom{14}{4}.$$

Next subtract those with $x_1 \geq 6$. For such solutions let $z_1 = x_1 - 6 \geq 0$; then

$$z_1 + x_2 + x_3 + x_4 + x_5 = 4,$$

whose nonnegative solutions are

$$\binom{4+5-1}{5-1} = \binom{8}{4}.$$

Therefore the number of solutions with $0 \leq x_1 \leq 5$ is

$$\binom{14}{4} - \binom{8}{4} = 931.$$

Q.13

Q.13 16 points are chosen inside a 5×3 rectangle. Prove that two of these points lie within $\sqrt{2}$ of each other.

PF:

In any unit square, the farthest two points can be apart is the length of the diagonal:

$$\sqrt{1^2 + 1^2} = \sqrt{2}.$$

Thus any two points lying in the same unit square are at distance at most $\sqrt{2}$.

By the pigeonhole principle, at least one unit square contains at least two points. For that pair of points, their distance is at most $\sqrt{2}$.

Therefore among any 16 points in the 5×3 rectangle, two of them lie within distance $\sqrt{2}$ of each other.

`Q.E.D.`

Q.14

Q.14 Let (x_i, y_i), $i = 1, 2, 3, 4, 5$, be a set of five distinct points with integer coordinates in the xy plane. Show that the midpoint of the line joining at least one pair of these points has integers coordinates.

PF:

For each point (x_i, y_i), consider the parity of its coordinates and record the pair

$$(x_i \bmod 2,\ y_i \bmod 2).$$

Each coordinate is either 0 or 1 modulo 2, so there are only four possible types:

$$(0,0),\ (0,1),\ (1,0),\ (1,1).$$

With five points and only four types, the pigeonhole principle implies that two points, say (x_i, y_i) and (x_j, y_j) with $i \neq j$, have the same type. Thus

$$x_i \equiv x_j \pmod 2, \qquad y_i \equiv y_j \pmod 2.$$

Hence $x_i + x_j$ and $y_i + y_j$ are both even, so

$$\frac{x_i + x_j}{2}, \qquad \frac{y_i + y_j}{2}$$

are integers.

Therefore the midpoint

$$\left(\frac{x_i + x_j}{2}, \frac{y_i + y_j}{2}\right)$$

has integer coordinates.

`Q.E.D.`

Q.15

Q.15 Show that if p is a prime and k is an integer such that $1 \leq k \leq p-1$, then p divides $\binom{p}{k}$.

PF:

We have

$$\binom{p}{k} = \frac{p!}{k!(p-k)!}.$$

The factorial $p!$ contains the factor p exactly once.

Since $1 \le k \le p-1$, both k and $p-k$ are less than p, so all factors of $k!$ and $(p-k)!$ are $< p$; in particular,

$$p \nmid k!, \qquad p \nmid (p-k)!,$$

and therefore

$$\gcd\big(p,\ k!(p-k)!\big) = 1.$$

Thus the factor p in the numerator cannot be canceled with the denominator when the fraction is reduced. The reduced numerator still contains p, so p divides $\binom{p}{k}$.

Q.E.D.

Q.16

Q.16 Prove the hockeystick identity

$$\sum_{k=0}^{r} \binom{n+k}{k} = \binom{n+r+1}{r}$$

whenever n and r are positive integers,

(a) using a combinatorial argument

(b) using Pascal's identity.

PF:

1.

Using symmetry, $\binom{n+k}{k} = \binom{n+k}{n}$ and $\binom{n+r+1}{r} = \binom{n+r+1}{n+1}$, so we need to show

$$\sum_{k=0}^{r} \binom{n+k}{n} = \binom{n+r+1}{n+1}.$$

Interpret $\binom{n+r+1}{n+1}$ as the number of ways to choose $n+1$ elements from $\{1, 2, \ldots, n+r+1\}$.

In any such choice, let L be the largest chosen element.

Then L can be any of $n+1, n+2, \ldots, n+r+1$.

Write $L = n+k+1$ with $k = 0, 1, \ldots, r$.

Once L is fixed, the remaining n elements must be chosen from $\{1, \ldots, n+k\}$, which can be done in $\binom{n+k}{n}$ ways.

Different k give disjoint families of choices, and together they account for all choices of $n+1$ elements.

Hence

$$\sum_{k=0}^{r} \binom{n+k}{n} = \binom{n+r+1}{n+1},$$

which is equivalent to

$$\sum_{k=0}^{r} \binom{n+k}{k} = \binom{n+r+1}{r}.$$

2.

Let

$$S_r = \sum_{k=0}^{r} \binom{n+k}{k}.$$

For $r = 0$, we have $S_0 = \binom{n}{0} = 1 = \binom{n+1}{0}$, so the identity holds.

Assume for some $r-1 \geq 0$ that

$$S_{r-1} = \sum_{k=0}^{r-1} \binom{n+k}{k} = \binom{n+r}{r-1}.$$

Then

$$S_r = \sum_{k=0}^{r} \binom{n+k}{k} = S_{r-1} + \binom{n+r}{r} = \binom{n+r}{r-1} + \binom{n+r}{r}.$$

By Pascal's identity $\binom{m}{t-1} + \binom{m}{t} = \binom{m+1}{t}$ with $m = n+r$ and $t = r$, we get

$$\binom{n+r}{r-1} + \binom{n+r}{r} = \binom{n+r+1}{r}.$$

Thus $S_r = \binom{n+r+1}{r}$, and by induction

$$\sum_{k=0}^{r} \binom{n+k}{k} = \binom{n+r+1}{r}$$

for all $r \geq 0$.

`Q.E.D.`

Q.17

Q.17

Solve the recurrence relation

$$a_n = 2a_{n-1} + a_{n-2} - 2a_{n-3}$$

with initial conditions $a_0 = 1$, $a_1 = 0$, and $a_2 = 7$.

Sol:

Assume a solution of the form $a_n = r^n$. Substituting into $a_n = 2a_{n-1} + a_{n-2} - 2a_{n-3}$ gives the characteristic equation

$$r^3 - 2r^2 - r + 2 = 0.$$

Factoring,

$$r^3 - 2r^2 - r + 2 = (r-1)(r-2)(r+1),$$

so the roots are $1, 2, -1$. Hence the general solution is

$$a_n = A + B2^n + C(-1)^n.$$

Use the initial conditions:

For $n = 0$:

$$1 = a_0 = A + B + C.$$

For $n = 1$:

$$0 = a_1 = A + 2B - C.$$

For $n = 2$:

$$7 = a_2 = A + 4B + C.$$

Solving,

$$B = 2, \quad C = \frac{3}{2}, \quad A = -\frac{5}{2}.$$

Therefore

$$a_n = -\frac{5}{2} + 2 \cdot 2^n + \frac{3}{2}(-1)^n = 2^{n+1} + \frac{3(-1)^n - 5}{2}.$$

Q.18

Q.18

(a) Find all solutions of the recurrence relation $a_n = 2a_{n-1} + 2n^2$.

(b) Find the solution of the recurrence relation in part (a) with initial condition $a_1 = 4$.

Sol:

1. The recurrence is

$$a_n = 2a_{n-1} + 2n^2.$$

The associated homogeneous recurrence $a_n^{(h)} = 2a_{n-1}^{(h)}$ has general solution

$$a_n^{(h)} = C2^n.$$

For a particular solution, try $a_n^{(p)} = An^2 + Bn + D$. Then

$$a_n^{(p)} - 2a_{n-1}^{(p)} = (-A-2)n^2 + (4A-B)n + (-2A+2B-D).$$

Setting this equal to $2n^2$ gives

$$-A-2=2, \quad 4A-B=0, \quad -2A+2B-D=0,$$

so

$$A=-2, \quad B=-8, \quad D=-12.$$

Thus

$$a_n^{(p)} = -2n^2 - 8n - 12.$$

The general solution is

$$a_n = C2^n - 2n^2 - 8n - 12.$$

---

2. Using $a_1 = 4$:

$$4 = a_1 = C2^1 - 2 \cdot 1^2 - 8 \cdot 1 - 12 = 2C - 22,$$

so $2C = 26$ and $C = 13$. Therefore

$$a_n = 13 \cdot 2^n - 2n^2 - 8n - 12.$$

Q.19

Q.19 Denote by a_n the number of *ternary* strings (with elements $0, 1, 2$) of length n that contain either 00 or 11.

(1) Find a recurrence relation for a_n with initial conditions.

(2) Find a closed-form expression for a_n.

Sol:

1. Let b_n be the number of ternary strings of length n that avoid 00 and 11. Then $a_n = 3^n - b_n$.

   To find b_n, classify valid strings of length n by their last symbol. Let x_n, y_n, z_n be the numbers ending in $0, 1, 2$ respectively. From the allowed transitions,

$$x_n = y_{n-1} + z_{n-1}, \quad y_n = x_{n-1} + z_{n-1}, \quad z_n = x_{n-1} + y_{n-1} + z_{n-1} = b_{n-1}.$$

   Since $b_n = x_n + y_n + z_n$ and $z_{n-1} = b_{n-2}$, we obtain

$$b_n = 2b_{n-1} + b_{n-2} \quad (n \geq 2),$$

   with $b_0 = 1$, $b_1 = 3$ (so $b_2 = 7$). Therefore

$$a_n = 3^n - b_n, \quad a_1 = 0,\ a_2 = 2,$$

   and for $n \geq 3$,

$$a_n = 2a_{n-1} + a_{n-2} + 2 \cdot 3^{n-2}.$$

---

2. For b_n we solve $b_n = 2b_{n-1} + b_{n-2}$, $b_0 = 1$, $b_1 = 3$. The characteristic equation

$$r^2 - 2r - 1 = 0$$

has roots $r = 1 \pm \sqrt{2}$. Thus

$$b_n = \alpha(1+\sqrt{2})^n + \beta(1-\sqrt{2})^n.$$

Using $b_0 = 1$, $b_1 = 3$ gives

$$\alpha = \frac{\sqrt{2}+1}{2}, \quad \beta = \frac{1-\sqrt{2}}{2},$$

so

$$b_n = \frac{1}{2}(1+\sqrt{2})^{n+1} + \frac{1}{2}(1-\sqrt{2})^{n+1}.$$

Therefore

$$a_n = 3^n - b_n = 3^n - \frac{1}{2}\Big((1+\sqrt{2})^{n+1} + (1-\sqrt{2})^{n+1}\Big).$$

Q.20

Q.20 Let $S_n = \{1, 2, \ldots, n\}$ and let a_n denote the number of *non-empty* subsets of S_n that contain **no** two consecutive integers. Find a recurrence relation for a_n. Note that $a_0 = 0$ and $a_1 = 1$.

Sol:

Let b_n be the number of subsets of S_n containing no two consecutive integers, including the empty set. Then $b_n = a_n + 1$.

Divide the subsets counted by b_n into two cases:

- If n is not in the subset, we get any valid subset of S_{n-1}, giving b_{n-1} subsets.
- If n is in the subset, then $n-1$ cannot be, and removing n leaves any valid subset of S_{n-2}, giving b_{n-2} subsets.

Hence, for $n \geq 2$,

$$b_n = b_{n-1} + b_{n-2},$$

with $b_0 = 1$ (only $\varnothing$) and $b_1 = 2$ ($\varnothing, \{1\}$).

Since $b_n = a_n + 1$, we have

$$a_n + 1 = (a_{n-1} + 1) + (a_{n-2} + 1),$$

so for $n \geq 2$,

$$a_n = a_{n-1} + a_{n-2} + 1,$$

with initial conditions

$$a_0 = 0, \qquad a_1 = 1.$$

Q.21

Q.21 Use generating functions to prove Pascal's identity: $C(n,r) = C(n-1,r) + C(n-1,r-1)$ when n and r are positive integers with $r < n$. [Hint: Use the identity $(1+x)^n = (1+x)^{n-1} + x(1+x)^{n-1}$.]

PF:

From the binomial theorem,

$$(1+x)^n = \sum_{r=0}^{n} C(n,r)x^r.$$

Using the hinted identity,

$$(1+x)^n = (1+x)^{n-1} + x(1+x)^{n-1}.$$

Expand the right-hand side:

$$(1+x)^{n-1} = \sum_{r=0}^{n-1} C(n-1,r)x^r,$$

$$x(1+x)^{n-1} = x\sum_{r=0}^{n-1} C(n-1,r)x^r = \sum_{r=0}^{n-1} C(n-1,r)x^{r+1} = \sum_{r=1}^{n} C(n-1,r-1)x^r.$$

Hence

$$(1+x)^{n-1} + x(1+x)^{n-1} = \sum_{r=0}^{n-1} C(n-1,r)x^r + \sum_{r=1}^{n} C(n-1,r-1)x^r.$$

For $1 \le r \le n-1$, the coefficient of x^r on the right-hand side is

$$C(n-1,r) + C(n-1,r-1).$$

But on the left-hand side, $(1+x)^n$ has coefficient $C(n,r)$ at x^r.

Equating coefficients of x^r for $1 \le r \le n-1$ gives

$$C(n,r) = C(n-1,r) + C(n-1,r-1),$$

which is Pascal's identity.

Q.E.D.

Q.22

Q.22 Use generating functions to prove Vandermonde's identity:

$$C(m+n,r) = \sum_{k=0}^{r} C(m,r-k)C(n,k),$$

whenever m, n, and r are nonnegative integers with r not exceeding either m or n. [Hint: Look at the coefficient of x^r in both sides of $(1+x)^{m+n} = (1+x)^m(1+x)^n$.]

PF:

From the binomial theorem,

$$(1+x)^{m+n} = \sum_{r=0}^{m+n} C(m+n,r)x^r.$$

Also,

$$(1+x)^{m+n} = (1+x)^m(1+x)^n.$$

Expand each factor:

$$(1+x)^m = \sum_{i=0}^{m} C(m,i)x^i, \qquad (1+x)^n = \sum_{j=0}^{n} C(n,j)x^j.$$

Their product is

$$(1+x)^m(1+x)^n = \left(\sum_{i=0}^{m} C(m,i)x^i\right)\left(\sum_{j=0}^{n} C(n,j)x^j\right).$$

By the Cauchy product, the coefficient of x^r in this product is

$$\sum_{k=0}^{r} C(m,r-k)C(n,k).$$

But in $(1+x)^{m+n}$ the coefficient of x^r is $C(m+n,r)$. Equating the coefficients of x^r on both sides gives

$$C(m+n,r) = \sum_{k=0}^{r} C(m,r-k)C(n,k),$$

which is Vandermonde's identity.

`Q.E.D.`

Q.23

Q.23 Generating functions are very useful, for example, provide an approach to solving linear recurrence relations. Read pp. 537-548 of the textbook. [You do not need to write anything for this problem on your submitted assignment paper.]

void.